Gurugram University Scheme of Studies and Examination

Bachelor of Technology Semester 4

S.No.	Category	Course Code	Course Title	Hours per week			Credits	Marks for Sessional	Marks for End Term Examination	Total
				L	T	P				
1	PCC		Transmission and Distribution	3	1	0	3	30	70	100
2	PCC		Electrical Machine II	3	1	0	3	30	70	100
3	PCC		Power Electronics	3	1	0	3	30	70	100
4	PCC		Electronic Measurement and Instrumentation	3	0	0	3	30	70	100
5	PCC		Electric Engineering Materials	3	0	0	3	30	70	100
6	BSC		Probability Theory and Stochastic Processes	3	1	0	3	30	70	100
7	LC		Transmission and Distribution Lab	0	0	2	1	50	50	100
8	LC		Electrical Machine II Lab	0	0	2	1	50	50	100
9	LC		Power Electronics Lab	0	0	2	1	50	50	100
10	LC		Electronic Measurement and Instrumentation Lab	0	0	2	1	50	50	100
11	Non credit		Scientific and Technical Writing Skills*	2	0	0	-	30	70	100*
Total							**22**			**1000**

***Note:** The examination of the regular students will be conducted by the concerned college/Institute internally. Each student will be required to score a minimum of 40% marks to qualify in the paper. The marks will not be included in determining the percentage of marks obtained for the award of a degree.

NOTE: At the end of 4th semester each student has to undergo Practical Training of 4/6 weeks in an Industry/Institute/ Professional Organization/Research Laboratory/ training centre etc. and submit typed report along with a certificate from the organization and its evaluation shall be carried out in the 5th Semester.

TRANSMISSION AND DISTRIBUTION

<table>
<tr><td>Course Code</td><td colspan="5"></td></tr>
<tr><td>Category</td><td colspan="5">Professional Core Courses</td></tr>
<tr><td>Course title</td><td colspan="5">Transmission and Distribution</td></tr>
<tr><td rowspan="2">Scheme</td><td>L</td><td>T</td><td>P</td><td>Credits</td><td rowspan="2">Semester: IV</td></tr>
<tr><td>3</td><td>1</td><td>0</td><td>3</td></tr>
<tr><td>Class Work</td><td colspan="5">30 Marks</td></tr>
<tr><td>Exam</td><td colspan="5">70 Marks</td></tr>
<tr><td>Total</td><td colspan="5">100 Marks</td></tr>
<tr><td>Duration of Exam</td><td colspan="5">3Hrs</td></tr>
</table>

Note: The examiner will set nine questions in total. Question one will be compulsory. Question one will have seven parts of 2 marks each from all units, and the remaining eight questions of 14 marks each to be set by taking two questions from each unit. The students have to attempt five questions in total, the first being compulsory and selecting one from each unit.

Course Objective: The objectives of this course are as under:

1. To introduce the basic laws of Transmission and Distribution
2. To impart knowledge on the Structure and present-Day Scenario of a power system.
3. To impart knowledge on the concepts of transmission and distribution line parameters.
4. To impart knowledge on the concepts of mechanical design of transmission line.
5. To impart knowledge on the performance of transmission line.

Unit-I

INTRODUCTION: Evolution of Power Systems and Present-Day Scenario. Structure of a power system, Bulk Power Grids and Micro-grids, indoor and outdoor substations, equipment for substations, layout, auxiliary supply.
DISTRIBUTION SYSTEMS: Radial, ring mains and network distribution system, comparison of various types of ac and dc systems.

Unit-II

TRANSMISSION LINES: Calculation of line parameters, Ferranti effect, proximity effect.
PERFORMANCE OF LINES: models of short, medium and long transmission lines, performance of transmission lines, circle diagram, capacity of synchronous condenser, tuned lines, voltage control.

Unit-III

MECHANICAL DESIGN: Sag and stress calculations, effect of ice and wind, dampers.
INSULATORS: Types, insulating materials, voltage distribution over insulator string, equalizer ring.

Unit-IV

CABLES: Types of LV and HV cables, grading of cables, capacitance, ratings.
CORONA: Phenomenon, critical voltage, power loss, reduction in losses, radio-interference,
HVDC transmission – types of links, advantages and limitations.

Course Outcomes:
At the end of the course, students will demonstrate the ability to:

1. Understand the basic laws of Transmission and Distribution
2. Knowledge about the Structure and present-Day Scenario of a power system.
3. Analyses of transmission and distribution line parameters.
4. Understand mechanical design of transmission line with skin effect and proximity effect.
5. Understand the various cables and insulators gradings as well as ratings.
6. To know the performance of transmission line.

Text/Ref. Books:

1. Power System Engg: I. J. Nagrath and D. P. Kothari (TMH)
2. Electrical Power Systems: C. L. Wadhwa (New Age International Pvt Ltd)
3. Grainger and W. D. Stevenson, "Power System Analysis", McGraw Hill Education, 1994.
4. Elements of power system analysis: W. D. Stevenson (MGH)
5. Electric Power System: B. M. Weedy, John Wiley & Sons.
6. Transmission & Distribution of Electrical Engineering: H. Cotton.
7. Transmission & Distribution of Electrical Engineering: Westing House & Oxford Univ. Press, New Delhi.

ELECTRICAL MACHINES-II

<table>
<tr><td>Course Code</td><td colspan="5"></td></tr>
<tr><td>Category</td><td colspan="5">Professional Core Courses</td></tr>
<tr><td>Course title</td><td colspan="5">Electrical Machines II</td></tr>
<tr><td rowspan="2">Scheme</td><td>L</td><td>T</td><td>P</td><td>Credits</td><td rowspan="2">Semester: IV</td></tr>
<tr><td>3</td><td>1</td><td>0</td><td>3</td></tr>
<tr><td>Class Work</td><td colspan="5">30 Marks</td></tr>
<tr><td>Exam</td><td colspan="5">70 Marks</td></tr>
<tr><td>Total</td><td colspan="5">100 Marks</td></tr>
<tr><td>Duration of Exam</td><td colspan="5">3Hrs</td></tr>
</table>

Note: The examiner will set nine questions in total. Question one will be compulsory. Question one will have seven parts of 2 marks each from all units, and the remaining eight questions of 14 marks each to be set by taking two questions from each unit. The students have to attempt five questions in total, the first being compulsory and selecting one from each unit.

Course Objective: The objectives of this course are as under:
1. To introduce the basic concepts of rotating magnetic fields.
2. To impart knowledge on the operation of ac machines.
3. To impart knowledge on the performance characteristics of ac machines
4. To impart knowledge on speed and torque characteristics of ac machine.
5. To impart knowledge on the motoring, generating and braking mode of ac machines.

Unit-I
Poly-phase Induction Motor: Constructional features, Principal of operation, production of rotating magnetic field, induction motor action, torque production, testing, development of equivalent circuit, performance characteristics, circle diagram, starting methods, double cage and deep bar motors.

Unit-II
Poly-phase Induction Motor: Methods of speed control - stator voltage control, stator resistance control, frequency control, rotor resistance control, slip power recovery control Induction Generator: Principle of operation, types and applications. Single Phase Induction motors: Double revolving field theory, cross field theory, different types of single-phase induction motors, circuit model of single-phase induction motor.

Unit-III
Synchronous Generator: Principle, construction of cylindrical rotor and salient pole machines, winding, EMF equation, Armature reaction, testing, model of the machine, regulation – synchronous reactance method, Potier triangle method. Output power equation, power angle curve.

Unit-IV
Three Phase Synchronous Generators: Transient and sub-transient reactance, synchronization, parallel operation.
Synchronous Motor: Principles of synchronous motor, power angle curve, V-curve, starting, damper winding, synchronous condenser, applications.

Course Outcomes: At the end of this course, students will demonstrate the ability to:
1. Understand the concepts of rotating magnetic fields.
2. Overview of construction of ac machines.
3. Understand the operation of ac machines.
4. Analyse performance characteristics of ac machines.
5. Impart knowledge on speed and torque characteristics of ac machine.
6. Prepare the students to have a basic knowledge about motoring, generating and braking mode of ac machines

Text/ reference books:

1. Principle of Electrical Machines, V K Mehta, Rohit Mehta, S Chand
2. Electric Machines, Ashfaq Hussain, Dhanpat Rai
3. Electric Machines: I.J.Nagrath and D.P. Kothari, TMH, New Delhi.
4. Generalized theory of Electrical Machines: P.S. Bhimbra(Khanna Pub.)
5. Electric Machinery, Fitzgerald and Kingsley, MGH.

POWER ELECTRONICS

<table>
<tr><td>Course Code</td><td colspan="5"></td></tr>
<tr><td>Category</td><td colspan="5">Professional Core Courses</td></tr>
<tr><td>Course title</td><td colspan="5">Power Electronics</td></tr>
<tr><td rowspan="2">Scheme</td><td>L</td><td>T</td><td>P</td><td>Credits</td><td rowspan="2">Semester: IV</td></tr>
<tr><td>3</td><td>1</td><td>0</td><td>3</td></tr>
<tr><td>Class Work</td><td colspan="5">30 Marks</td></tr>
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Course Objective: The objectives of this course are as under:
1. To introduce the basic of power switching device.
2. To impart knowledge on the concepts of Rectifiers and regulators.
3. To impart knowledge on the concepts of converters.
4. To impart knowledge on the concepts of inverter.
5. To impart knowledge on the concepts of cycloconverter.

Unit-I

Power switching devices
Diode, Thyristor, MOSFET, IGBT: I-V Characteristics; Protections, series and parallel connections, Firing circuit for thyristor; Voltage and current commutation of a thyristor; pulse transformer and opto-coupler.
AC REGULATORS: Types of regulators, equation of load current, calculation of extinction angle, output voltage equation, harmonics in load voltage.

Unit-II

Thyristor rectifiers
Single-phase half-wave and full-wave rectifiers, Single-phase full-bridge thyristor rectifier with R- load and highly inductive load; Three phase full-bridge thyristor rectifier with R-load and highly inductive load; Input and output wave shape and power factor.
DC-DC buck converter
Elementary chopper with an active switch and diode, concepts of duty ratio and average voltage, power circuit of a buck converter, analysis and waveforms at steady state, duty ratio control of output voltage.

Unit-III

DC-DC boost converter:
Power circuit of a boost converter, analysis and waveforms at steady state, relation between duty ratio and average output voltage.
Single-phase voltage source inverter:
Power circuit of single-phase voltage source inverter, switch states and instantaneous output voltage, square wave operation of the inverter, concept of average voltage over a switching cycle, bipolar sinusoidal modulation and unipolar sinusoidal modulation, modulation index and output voltage

Unit-IV

Three-phase voltage source inverter
Power circuit of a three-phase voltage source inverter, switch states, instantaneous output voltages, average output voltages over a sub cycle, three-phase sinusoidal modulation.
Cycloconverters:
Basic principle of frequency conversion, types of cycloconverter, non-circulating and circulating types of cycloconverters

Course Outcomes: At the end of this course students will demonstrate the ability to;
1. Understand the differences between signal level and power level devices.
2. Understand working of AC regulators.
3. Analyse controlled rectifier circuits.
4. Analyse the operation of DC-DC choppers.
5. Analyse the operation of voltage source inverters.

6. Analyse cycloconverters.

Text/References Books: -

1. M. H. Rashid, “Power electronics: circuits, devices, and applications”, Pearson Education India, 2009.
2. N. Mohan and T. M. Undeland, “Power Electronics: Converters, Applications and Design”, John Wiley & Sons, 2007.
3. R. W. Erickson and D. Maksimovic, “Fundamentals of Power Electronics”, Springer Science & Business Media, 2007.
4. L. Umanand, “Power Electronics: Essentials and Applications”, Wiley India, 2009.

ELECTRONIC MEASUREMENT AND INSTRUMENTATION

<table>
<tr><td>Course Code</td><td colspan="5"></td></tr>
<tr><td>Category</td><td colspan="5">Professional Core Courses</td></tr>
<tr><td>Course title</td><td colspan="5">Electronic Measurement and Instrumentation</td></tr>
<tr><td rowspan="2">Scheme</td><td>L</td><td>T</td><td>P</td><td>Credits</td><td rowspan="2">Semester: IV</td></tr>
<tr><td>3</td><td>0</td><td>0</td><td>3</td></tr>
<tr><td>Class Work</td><td colspan="5">30 Marks</td></tr>
<tr><td>Exam</td><td colspan="5">70 Marks</td></tr>
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Course Objective: The objectives of this course are as under:
1. To introduce Electronics Instruments and Measurement providing an in-depth understanding of Measurement errors.
2. Digital Storage Oscilloscope, Function Generator and Analyzer, Display devices, Data acquisition systems and transducers.
3. To address the underlying concepts and methods behind Electronics measurements.
4. To introduce signal conditioning.

Unit-I

OSCILLOSCOPE: Block diagram, study of various stages in brief, high frequency CRO considerations. Sampling and storage oscilloscope.
GENERATION and ANALYSIS OF WAVEFORMS: Block diagram of pulse generators, signal generators, function generators wave analysers, distortion analysers, spectrum analyser, Harmonic analyser, introduction to power analyser.

Unit-II

ELECTRONIC INSTRUMENTS: Instruments for measurement of voltage, current and other circuit parameters, Q meters, R.F. Power measurements, introduction to digital meters.
FREQUENCY and TIME MEASUREMENT: Study of decade counting Assembly (DCA), frequency measurements, period measurements, Universal counter, Introduction to digital meters.

Unit-III

DISPLAY DEVICES: Nixie tubes, LED's LCD's, discharge devices.
TRANSDUCERS: Classification, Transducers of types: RLC photocell, thermocouples etc. basic schemes of measurement of displacement, velocity, acceleration, strain, pressure, liquid level and temperature.

Unit-IV

INTRODUCTION TO SIGNAL CONDITIONING:
DC signal conditioning system, AC signal conditioning system, data acquisition and conversion system

Course Outcome:
1. Analyze the performance characteristics of each instrument
2. Illustrate basic meters such as voltmeters and ammeters.
3. Explain about different types of signal analyzers.
4. Explain the basic features of oscilloscope and different types of oscilloscopes
5. Identify the various parameters that are measurable in electronic instrumentation.
6. Employ appropriate instruments to measure given sets of parameters.

Text book:
1. A course in Electrical & Electronics Measurements & Instrumentation: A. K. Sawhney; Dhanpat Rai & Sons.

Reference books.
1. Electronics Instrumentation & Measurement Techniques: Cooper; PHI.

ELECTRICAL ENGINEERING MATERIALS

<table>
<tr><td>Course Code</td><td colspan="5"></td></tr>
<tr><td>Category</td><td colspan="5">Professional Core Courses</td></tr>
<tr><td>Course title</td><td colspan="5">Electrical Engineering Materials</td></tr>
<tr><td rowspan="2">Scheme</td><td>L</td><td>T</td><td>P</td><td>Credits</td><td rowspan="2">Semester: IV</td></tr>
<tr><td>3</td><td>0</td><td>0</td><td>3</td></tr>
<tr><td>Class Work</td><td colspan="5">30 Marks</td></tr>
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Course Objective: The objectives of this course are as under:
1. To provide the Information of metal.
2. This course is an introduction to dielectric properties of material
3. This course is an introduction to magnetic properties of material
4. The goal is to provide a basic understanding of semiconductors.

Unit-I

Conductivity of Metal: Introduction, factors affecting the resistivity of electrical materials, motion of an electron in an electric field, Equation of motion of an electron, current carried by electrons, mobility, thermionic emission, photo electric emission, field emission, effect of temperature on electrical conductivity of metals, electrical conducting materials, thermal properties, thermal conductivity of metals, thermoelectric effects.

Unit-II

Dielectric Properties: Introduction, effect of a dielectric on the behavior of a capacitor, polarization, the dielectric constant of monatomic gases, dielectric losses, significance of the loss tangent, frequency and temperature dependence of the dielectric constant, dielectric properties of polymeric system, ionic conductivity in insulators, insulating materials, ferroelectricity, piezoelectricity

Unit-III

Magnetic properties of Materials: Introduction, Classification of magnetic materials, diamagnetism, paramagnetism, ferromagnetism, magnetization curve, the hysteresis loop, factors affecting permeability and hysteresis loss, common magnetic materials, magnetic resonance.

Unit-IV

Semiconductors: energy band in solids, conductors, semiconductors and insulators, types of semiconductors, Intrinsic semiconductors, impurity type semiconductor, diffusion, the Einstein relation, hall effect, thermal conductivity of semiconductors, electrical conductivity of doped materials.

Course outcome:
After the completion of the course, the students will be able to:
1. Learn the basics of metal.
2. Learn the basics of conductivity.
3. Realize the dielectric properties of materials.
4. Explain the importance of magnetic properties.
5. Explain the behavior of conductivity of metals and
6. Classify semiconductor material.

Text book:

1. Bhadra Prasad Pokharel and Nava Raj Karki, "Electrical Engineering Materials", Sigma offset Press, Kamaladi, Kathmandu, Nepal,2004.
2. R.C. Jaeger," Introduction to Microelectronic Fabrication- Volume IV", Addison Wesley publishing Company, Inc., 1988.
3. Kasap.S.O, Principles of electrical engineering materials and devices, McGraw Hill, NewYork,2000.
4. R. A. Colcaser and S. Diehl-Nagle," Materials and Devices for Electrical Engineers and Physicists, McGraw-Hill, New York, 1985.

Reference books

1. C.S.Indulkar and S. Thiruvengadam, S., “An Introduction to Electrical Engineering”
2. Kenneth G. Budinski,, “Engineering Materials: Prentice Hall of India, New Delhi

PROBABILITY THEORY AND STOCHASTIC PROCESSES

<table>
<tr><td>Course Code</td><td colspan="5"></td></tr>
<tr><td>Category</td><td colspan="5">Basic Science Courses</td></tr>
<tr><td>Course title</td><td colspan="5">Probability Theory and Stochastic Processes</td></tr>
<tr><td rowspan="2">Scheme</td><td>L</td><td>T</td><td>P</td><td>Credits</td><td rowspan="2">Semester: IV</td></tr>
<tr><td>3</td><td>1</td><td>0</td><td>3</td></tr>
<tr><td>Class Work</td><td colspan="5">30 Marks</td></tr>
<tr><td>Exam</td><td colspan="5">70 Marks</td></tr>
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Course Objective: The objectives of this course are as under:

1. To introduce the probability theory and random processes and illustrate these concepts with engineering applications.
2. To introduce random variables.
3. The course introduces the concept of Stochastic Processes.
4. To understand regression analysis.

Unit-I

Sets and set operations; Probability space; Conditional probability and Bayes theorem; Combinatorial probability and sampling models. Discrete random variables, probability mass function, probability distribution function, example random variables and distributions; Continuous random variables, probability density function, probability distribution function, example distributions

Unit-II

Joint distributions, functions of one and two random variables, moments of random variables; Conditional distribution, densities and moments; Characteristic functions of a random variable; Markov, Chebyshev and Chernoff bounds;

Unit-III

Random sequences and modes of convergence (everywhere, almost everywhere, probability, distribution and mean square); Limit theorems; Strong and weak laws of large numbers, central limit theorem. Random process. Stationary processes. Mean and covariance functions. Ergodicity. Transmission of random process through LTI. Power spectral density.

Unit-IV

Regression analysis (linear and non-linear), Confidence intervals, Hypothesis testing, Error analysis

Course Outcomes:

1. Develop understanding of basics of probability theory.
2. Understand random variables.
3. Identify different distribution functions and their relevance.
4. Apply the concepts of probability theory to different problems.
5. Extract parameters of a stochastic process and use them for process characterization.
6. Apply regression analysis.

Text/Reference Books:

1. H. Stark and J. Woods, `Probability and Random Processes with Applications to Signal Processing," Third Edition, Pearson Education
2. A.Papoulis and S. Unnikrishnan Pillai, ``Probability, Random Variables and Stochastic Processes," Fourth Edition, McGraw Hill.
3. K. L. Chung, Introduction to Probability Theory with Stochastic Processes, Springer International
4. P. G. Hoel, S. C. Port and C. J. Stone, Introduction to Probability, UBS Publishers,
5. S. Ross, Introduction to Stochastic Models, Harcourt Asia, Academic Press

TRANSMISSION AND DISTRIBUTION LABORATORY

<table>
<tr><td>Course Code</td><td colspan="5"></td></tr>
<tr><td>Category</td><td colspan="5">Laboratory Courses</td></tr>
<tr><td>Course title</td><td colspan="5">Transmission and Distribution Laboratory</td></tr>
<tr><td rowspan="2">Scheme</td><td>L</td><td>T</td><td>P</td><td>Credits</td><td rowspan="2">Semester: IV</td></tr>
<tr><td>0</td><td>0</td><td>2</td><td>1</td></tr>
<tr><td>Class Work</td><td colspan="5">50 Marks</td></tr>
<tr><td>Exam</td><td colspan="5">50 Marks</td></tr>
<tr><td>Total</td><td colspan="5">100 Marks</td></tr>
<tr><td>Duration of Exam</td><td colspan="5">3Hrs</td></tr>
</table>

Notes:

(i) At least 10 experiments are to be performed by students in the semester.

(ii) At least 7 experiments should be performed from the list, remaining three experiments may either be performed from the above list or designed and set by the concerned institution as per the scope of the syllabus

LIST OF EXPERIMENTS:

1. To study the Power System blocks in MATLAB.
2. To design short and long transmission line using MATLAB.
3. To study and calculate the transmission line parameters.
4. To study the corona loss in power distribution system.
5. To study the proximity and skin effect.
6. To find ABCD parameters of a model of transmission line.
7. To study performance of a transmission line under no load condition and under load at different power factors.
8. To observe the Ferranti effect in a model of transmission line.
9. To study performance characteristics of typical DC distribution system in radial and ring main configuration.
10. 10.To study mechanical design of transmission line.

Lab Outcomes:

At the end of the lab, students will demonstrate the ability to:

1. Understand the basic laws of Transmission and Distribution
2. Knowledge about the Structure and present-Day Scenario of a power system.
3. Analyses of transmission and distribution line parameters.
4. Understand mechanical design of transmission line with skin effect and proximity effect.
5. Understand the various cables and insulators gradings as well as ratings.
6. To know the performance of transmission line.

ELECTRICAL MACHINE II LABORATORY

<table>
<tr><td>Course Code</td><td colspan="5"></td></tr>
<tr><td>Category</td><td colspan="5">Laboratory Courses</td></tr>
<tr><td>Course title</td><td colspan="5">Electrical Machine II Laboratory</td></tr>
<tr><td rowspan="2">Scheme</td><td>L</td><td>T</td><td>P</td><td>Credits</td><td rowspan="2">Semester: IV</td></tr>
<tr><td>0</td><td>0</td><td>2</td><td>1</td></tr>
<tr><td>Class Work</td><td colspan="5">50 Marks</td></tr>
<tr><td>Exam</td><td colspan="5">50 Marks</td></tr>
<tr><td>Total</td><td colspan="5">100 Marks</td></tr>
<tr><td>Duration of Exam</td><td colspan="5">3Hrs</td></tr>
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Notes:

(i) At least 10 experiments are to be performed by students in the semester.

(ii) At least 7 experiments should be performed from the list, remaining three experiments may either be performed from the above list or designed and set by the concerned institution as per the scope of the syllabus

LIST OF EXPERIMENTS:

1. To perform the open circuit test and block rotor test on 3 phase induction motor and draw the circle diagram.
2. To study the speed control of induction motor by rotor resistance control.
3. To conduct the load test to determine the performance characteristics of the I.M.
4. To compute the torque v/s speed characteristics for various stator voltages.
5. To perform the open circuit test and block rotor test on single-phase induction motor and determine equivalent circuit parameters.
6. To perform O.C. test on synchronous generator and determine the full load regulation of a three phase synchronous generator by synchronous impedance method.
7. To Study and Measure Synchronous Impedance and Short circuit ratio of Synchronous Generator .
8. Study of Power (Load) sharing between two Three Phase alternators in parallel operation Condition.
9. To plot V- Curve of synchronous motor.
10. Synchronization of two Three Phase Alternators by
    a) Synchroscope Method
    b) Three dark lamp Method
    c) Two bright one dark lamp Method
11. Determination of sequence impedances of synchronous machine for various stator voltages.

Note:

1. Each laboratory group shall not be more than about 20 students.
2. To allow fair opportunity of practical hands-on experience to each student, each experiment may either done by each student individually or in group of not more than 3-4 students. Larger groups be strictly discouraged/disallowed.

Lab Outcomes:

At the end of this lab, students will demonstrate the ability to:

1. Understand the concepts of rotating magnetic fields.
2. Overview of construction of ac machines.
3. Understand the operation of ac machines.
4. Analyse performance characteristics of ac machines.
5. Impart knowledge on speed and torque characteristics of ac machine.

POWER ELECTRONICS LABORATORY

<table>
<tr><td>Course Code</td><td colspan="5"></td></tr>
<tr><td>Category</td><td colspan="5">Laboratory Courses</td></tr>
<tr><td>Course title</td><td colspan="5">Power Electronics Laboratory</td></tr>
<tr><td rowspan="2">Scheme</td><td>L</td><td>T</td><td>P</td><td>Credits</td><td rowspan="2">Semester: IV</td></tr>
<tr><td>0</td><td>0</td><td>2</td><td>1</td></tr>
<tr><td>Class Work</td><td colspan="5">50 Marks</td></tr>
<tr><td>Exam</td><td colspan="5">50 Marks</td></tr>
<tr><td>Total</td><td colspan="5">100 Marks</td></tr>
<tr><td>Duration of Exam</td><td colspan="5">3Hrs</td></tr>
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Notes:

(i) At least 10 experiments are to be performed by students in the semester.

(ii) At least 7 experiments should be performed from the list, remaining three experiments may either be performed from the above list or designed and set by the concerned institution as per the scope of the syllabus

LIST OF EXPERIMENTS

1. To Study Static Characteristics of Power Diode and Thyristor and to study reverse recovery of Power Diode and Thyristor.
2. To Study Characteristics of IGBT and MOSFET.
3. To study R, RC and UJT firing Circuit.
4. To Study of Pulse transformer and optocoupler technique
5. To Study of SCR Communication Technique Class, A-E.
6. To Study of AC voltage Regulator.
7. To control speed of small motor using Single Phase Half wave and Full wave fully controlled Converter
8. To control speed of a small DC motor using MOSFET based Chopper with output voltage control technique
9. To Study of Mc Murray - Bed ford Half and Full Bridge Inverter
10. To control speed of small AC induction motor using Single Phase non circulating type bridge by frequency conversion.
11. To Study single phase cycloconverter.

Note:

1. Each laboratory group shall not be more than about 20 students.
2. To allow fair opportunity of practical hands-on experience to each student, each experiment may either done by each student individually or in group of not more than 3-4 students. Larger groups be strictly discouraged/disallowed.

Lab Outcomes:

At the end of this lab, students will demonstrate the ability to;

1. Understand the differences between signal level and power level devices.
2. Understand working of AC regulators.
3. Analyse controlled rectifier circuits.
4. Analyse the operation of DC-DC choppers.
5. Analyse the operation of voltage source inverters.
6. Analyse cycloconverters/.

ELECTRONIC MEASUREMENT AND INSTRUMENTATION LABORATORY

<table>
<tr><td>Course Code</td><td colspan="5"></td></tr>
<tr><td>Category</td><td colspan="5">Laboratory Courses</td></tr>
<tr><td>Course title</td><td colspan="5">Electronic Measurement and Instrumentation Laboratory</td></tr>
<tr><td rowspan="2">Scheme</td><td>L</td><td>T</td><td>P</td><td>Credits</td><td rowspan="2">Semester: IV</td></tr>
<tr><td>0</td><td>0</td><td>2</td><td>1</td></tr>
<tr><td>Class Work</td><td colspan="5">50 Marks</td></tr>
<tr><td>Exam</td><td colspan="5">50 Marks</td></tr>
<tr><td>Total</td><td colspan="5">100 Marks</td></tr>
<tr><td>Duration of Exam</td><td colspan="5">3Hrs</td></tr>
</table>

Notes:
(i) At least 10 experiments are to be performed by students in the semester.
(ii) At least 7 experiments should be performed from the list, remaining three experiments may either be performed from the above list or designed and set by the concerned institution as per the scope of the syllabus.
(iii) Group of students for practical should be 15 to 20 in number.

LIST OF EXPERIMENTS
1. Study blocks wise construction of an analog oscilloscope and Function generator.
2. Study blocks wise construction of a Multimeter and frequency counter.
3. Study Measurement of different components and parameters like Q of a coil etc using LCRQ meter.
4. Study of distortion factor meter and determination of the % distortion of the given oscillator
5. Determine output characteristics of a LVDT and Measure displacement using LVDT
6. Study characteristics of temperature transducer like Thermocouple, Thermistor and RTD with implementation of a small project using signal conditioning circuits like instrumentation amplifier.
7. Measurement of Strain using Strain Guage.
8. To study differential pressure transducer and signal conditioning of output signal.
9. Measurement of level using capacitive transducer.
10. Study of Distance measurement using ultrasonic transducer.

Note:
1. Each laboratory group shall not be more than about 20 students.
2. To allow fair opportunity of practical hands-on experience to each student, each experiment may either done by each student individually or in group of not more than 3-4 students. Larger groups be strictly discouraged/disallowed.

Lab Outcome:
At the end of this lab, students will demonstrate the ability to;
1. Analyze the performance characteristics of each instrument
2. Illustrate basic meters such as voltmeters and ammeters.
3. Explain about different types of signal analyzers.
4. Explain the basic features of oscilloscope and different types of oscilloscopes
5. Identify the various parameters that are measurable in electronic instrumentation.
6. Employ appropriate instruments to measure given sets of parameters.

SCIENTIFIC AND TECHNICAL WRITING SKILLS

<table>
<tr><td>Course Code</td><td colspan="5"></td></tr>
<tr><td>Category</td><td colspan="5">Non-Credit</td></tr>
<tr><td>Course title</td><td colspan="5">Scientific and Technical Writing Skills</td></tr>
<tr><td rowspan="2">Scheme</td><td>L</td><td>T</td><td>P</td><td>Credits</td><td rowspan="2">Semester: IV</td></tr>
<tr><td>2</td><td>0</td><td>0</td><td>0</td></tr>
<tr><td>Class Work</td><td colspan="5">30</td></tr>
<tr><td>Exam</td><td colspan="5">70</td></tr>
<tr><td>Total</td><td colspan="5">100</td></tr>
<tr><td>Duration of Exam</td><td colspan="5">3Hrs</td></tr>
</table>

Note: The examination of the regular students will be conducted by the concerned college/Institute internally. Each student will be required to score a minimum of 40% marks to qualify in the paper. The marks will not be included in determining the percentage of marks obtained for the award of a degree.

The following course content to conduct the activities is prescribed for the Scientific and Technical writing Skills Lab:

1. Activities on Writing Skills - Structure and presentation of different types of writing - letter writing/ Resume writing/ e-correspondence/ Technical report writing/ Portfolio writing - planning for writing - improving one's writing.
2. Activities on Presentation Skills - Oral presentations (individual and group) through JAM sessions/seminars/PPTs and written presentations through posters/ projects/ reports/ e-mails/ assignments etc.
3. Activities on Group Discussion and Interview Skills - Dynamics of group discussion, intervention, summarizing, modulation of voice, body language, relevance, fluency and organization of ideas and rubrics for evaluation- Concept and process, pre-interview planning, opening strategies, answering strategies, interview through tele-conference and video-conferencing and Mock Interviews.

Text references:

1. A Course Book of Advanced Communication Skills (ACS) Lab published by Universities Press, Hyderabad.
2. Technical Communication by Meenakshi Raman & Sangeeta Sharma, Oxford University Press 2009.
3. Advanced Communication Skills Laboratory Manual by Sudha Rani, D, Pearson Education 2011.
4. Technical Communication by Paul V. Anderson, 2007. Cengage Learning pvt. Ltd. New Delhi.
5. Business and Professional Communication: Keys for Workplace Excellence, Kelly M. Quintanilla & Shawn T. Wahl. Sage South Asia Edition. Sage Publications, 2011.
6. The Basics of Communication: A Relational Prespective, Stev Duck & David T. Mc Mahan. Sage South Asia Edition. Sage Publications, 2012.
7. English Vocabulary in Use series, Cambridge University Press 2008.
8. Management Shapers Series by Universities Press (India) Pvt Ltd., Himayatnagar, Hyderabad 2008.
9. Handbook for Technical Communication by David A. McMurrey & Joanne Buckley, 2012. Cengage Learning.
10. Communication Skills by Leena Sen, PHI Learning Pvt Ltd., New Delhi, 2009.
11. Handbook for Technical Writing by David A McMurrey & Joanne Buckely CENGAGE Learning 2008.
12. Job Hunting by Colm Downes, Cambridge University Press 2008.
13. Master Public Speaking by Anne Nicholls, JAICO Publishing House, 2006.
14. English for Technical Communication for Engineering Students, Aysha Vishwamohan, Tata Mc graw Hill 2009.
15. Books on TOFEL/ GRE/ GMAT/ CAT/ IELTS by Barron's/ DELTA/ Cambridge University Press.
16. International English for Call Contres by Barry Tomalin and Suhashini Thomas, Macmillan Publishers, 2009.

Mini Project: As a part of Internal Evaluation

1. Seminar/ Professional Presentation
2. A Report on the same has to be prepared and presented.
    - Teachers may use their discretion to choose topics relevant and suitable to the needs of students.
    - Not more than two students to work on each mini project.
    - Students may be assessed by their performance both in oral presentation and written report.